Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims

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Abstract

Large language models can draft medication alerts whose fluency exceeds the evidence they cite. This paper studies one workflow: pharmacogenomic and genomic medication-alert text. I implement a synthetic pre-display evidence gate with three checks: citation/guideline support, population fit, and endpoint/actionability. Stage 1 does not call an LLM or parse free text; it tests a deterministic controller over structured annotations. In 33 author-designed synthetic cases, the controller blocked or narrowed all 23 designed overclaim archetypes, allowed all 10 bounded alerts, and produced no inappropriate denials. Action conformance was 33/33; primary-gate conformance was 31/33. These are specification-conformance results, not clinical safety, generalization, or patient-benefit evidence.

Keywords: clinical decision support; large language models; pharmacogenomics; evidence overclaiming; model evaluation

Data and Code Availability The Stage 1 case bank is synthetic and contains no real patient records, protected health information, or private genomic data. For anonymous venue review, code and synthetic data should be supplied as an anonymized supplemental archive; the deanonymized public repository may be disclosed after acceptance or in course-facing materials.

Institutional Review Board (IRB) Stage 1 uses only author-designed synthetic cases and public guideline/database names, and therefore does not require IRB approval. Any Stage 2 use of clinical cases or expert adjudication would require appropriate local governance.

1. Introduction

Healthcare LLMs can make evidentiary overreach look syntactically clean. A medication alert may cite a real guideline while turning a bounded pharmacogenomic recommendation into a universal instruction, a population-specific association into a transported claim, or a surrogate endpoint into patient-outcome benefit. The gap matters because fluency can hide what evidence does, and does not, authorize. Pharmacogenomic medication alerts are a useful testbed because CPIC and ClinPGx already distinguish genotype, phenotype, evidence level, and prescribing recommendation (Clinical Pharmacogenetics Implementation Consortium, 2026; ClinPGx, 2026a).

2. Evidence Boundary

Healthcare AI evaluation should distinguish implementation consistency from patient-outcome benefit. Byrne et al. emphasize that outcome claims require pragmatic clinical evaluation (Byrne et al., 2024), while Stegenga motivates caution when evidentiary pipelines make interventions appear stronger than they are (Stegenga, 2018). FDA–NIH BEST materials provide endpoint vocabulary: biomarkers, surrogate endpoints, validated surrogates, and clinical outcomes are not interchangeable (U.S. Food and Drug Administration and National Institutes of Health, 2016). Pharmacogenomics adds a positive control: CPIC-style resources can support bounded prescribing language, including CYP2C19–clopidogrel and DPYD–fluoropyrimidine examples (Lee et al., 2022; Amstutz et al., 2018), but do not automatically support claims that an LLM improves outcomes, that an association is causal, or that a result transports across populations without recalibration.

3. Methods

The controller is a deterministic pre-display gate, not an autonomous clinical agent. It receives a synthetic candidate alert plus structured evidence annotations assigned in the case bank; it does not parse draft text, call an LLM, verify citations, or extract EHR/population features. The fixed precedence order is citation/guideline support, population fit, then endpoint/actionability. It returns one of five permissions: allow, narrow, abstain for population fit, abstain for citation support, or deny unsupported action.

4. Stage 1 Scaffold

The case bank contains 33 author-designed synthetic cases: 10 bounded aligned alerts and 23 overclaim archetypes involving universal action language, unsupported genetics, population mismatch, citation weakness, variant uncertainty, obsolete/context-shifted evidence, and three isolating cases. The 23/10 split is stress-test coverage, not a clinical base rate. Each case has expected labels, but the controller does not read them. The three checks are summarized in Table 1.

Table 2 reports specification-conformance counts only: no designed overclaim reached allow unchanged, all bounded alerts passed, and no bounded alert was denied too strongly. The ablation now includes three isolating cases: a fabricated CPIC addendum attached to an otherwise actionable claim, a real guideline-supported claim transported to a mismatched population, and a stale but real guideline used as current evidence. Table 3 shows that disabling each gate allows at least one designed overclaim to pass unchanged, so the gates are not merely duplicating endpoint/actionability. Removing citation/guideline support allows 2/23 overclaims through unchanged; removing population fit allows 1/23 through; removing endpoint/actionability allows 6/23 through. The result supports a necessity chain for the three gates in this synthetic case bank.

PGX19 and PGX24 remain negative results: both are action-level successes but primary-gate mismatches because citation/guideline support fires before the gate that motivated the case design. This exposes controller precedence rather than hiding it inside a perfect conformance score.

5. Evaluation Plan

The independent Stage 2 evaluation should compare ungated LLM drafts with gated drafts on independently authored cases. Reviewers should be blinded to arm when feasible and classify overclaiming, inappropriate denial, preserved clinician authority, population-fit concerns, citation support, and error category.

6. Ethics and Limitations

The governance value of the gate is transparency. The controller is deterministic, auditable, and intentionally non-agentic: each alert can show which evidence feature triggered allow, narrow, abstain, or deny. Fairness and bias are central in pharmacogenomics. HLA-B*15:02 risk language is ancestry-conditional, founder-variant evidence may not transport to general US populations, and biobank results can misfit underrepresented or recently admixed groups (ClinPGx, 2026b; National Center for Biotechnology Information, 2026; Genome Aggregation Database Consortium, 2026; National Institutes of Health, 2026; Genome India Project, 2026; Psifas, 2026). Cases PGX09, PGX10, PGX25, PGX26, and PGX27 treat population fit as an equity constraint. PGX24 marks a high-risk boundary: a polygenic score should not be used to deny opioid or stimulant therapy without stronger evidence and governance. Stage 1 contains no PHI and makes no diagnosis or treatment recommendation; deployed clinical decision support would still need accountability for extraction, citation verification, thresholds, overrides, and audit logging. The key limitation is that results are synthetic and author-designed. They show rule conformance, not generalization, and assume structured annotations rather than extracting them from live LLM output or EHR context.

7. Conclusion

This small deterministic evidence gate separates bounded guideline support from unsupported action, population transport, and cita-

Table 1: Evidence-gate roles.

Gate	Question	Permitted effect
Citation	Is the cited source real, current, relevant, and strong enough?	Deny fabricated citations; abstain on uncertain support.
Population	Does source evidence plausibly fit the target patient or population?	Abstain from transport.
Endpoint	Does evidence support medication action, not only association?	Allow bounded alerts; narrow overstrong claims.

Table 2: Stage 1 specification-conformance counts.

Result	Value
Overclaim archetypes allowed unchanged	0/23
Designed overclaim archetypes blocked or narrowed	23/23
Bounded alerts allowed	10/10
Action conformance	33/33
Primary-gate conformance	31/33
Inappropriate denial	0/10

Table 3: One-gate ablation on the synthetic case bank.

Disabled gate	Allowed	Action Δ	Gate Δ
Citation/guideline	2	7	12
Population fit	1	5	5
Endpoint/actionability	6	6	6

tion overreach. It is a Findings-track design and evaluation-plan artifact with a reference implementation.

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